

Register Number

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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Computer Science and Engineering

Continuous Assessment Test – II

Regulations – R2021

Degree & Branch	BE - CSE				Semester	6
Subject Code & Name	UCS2604 - PRINCIPLES OF MACHINE LEARNING					
Academic Year	2023-2024 ODD /EVEN	Batch	2021-2025	Date	26.04.2024	FN / AN
Time: 08:15 - 09:45 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1	Explain the basic concepts of machine learning (K2)
CO2	Apply supervised algorithms for different classification problems (K3)
CO3	Explain the need for ensemble methods (K2)
CO4	Apply unsupervised and reinforcement learning techniques to various problems (K3)
CO5	Apply dimensionality reduction and optimization techniques (K3)

Part – A (4 × 2 = 8 Marks)

		KL	CO	PI
1.	Make use of the entropy metric and compute the impurity value for the following data: [5+, 5-], [0+, 10-], [8+, 2-] Write your inference about the purity of the data points from the calculated entropy values.	K3	CO3	1.1.1
2.	Compare Markov Decision Process with the traditional decision process in Reinforcement Learning (RL).	K2	CO4	1.4.1
3.	Make use of a suitable real-time scenario and describe the concept of unsupervised learning.	K3	CO4	1.4.1
4.	Compare Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) in handling the “curse of dimensionality”.	K2	CO5	1.4.1

Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	Consider the following dataset and apply a probabilistic model using Baye's Theorem to classify the species as 'M' or 'H'	K3	CO3	1.1.1 2.1.3 13.3.1

Sl. No.	Color	Legs	Height	Smelly	Species			
1	White	3	Short	Yes	M			
2	Green	2	Tall	No	M			
3	Green	3	Short	Yes	M			
4	White	3	Short	Yes	M			
5	Green	2	Short	No	H			
6	White	2	Tall	No	H			
7	White	2	Tall	No	H			
8	White	2	Short	Yes	H			
Using the above data, identify the species of an entity with the following attributes. $X = \{\text{Color}=\text{Green}, \text{Legs}=2, \text{Height}=\text{Tall}, \text{Smelly}=\text{No}\}$								
6.	Illustrate bagging and boosting techniques with suitable diagrams.					K2	CO3	1.4.1
7.	Compare Q learning with SARSA Algorithm in Reinforcement Learning.					K2	CO4	1.4.1

Part – C ($2 \times 12 = 24$ Marks)
(Any subdivisions should be either 7+5 or 8+4)

		KL	CO	PI
8.	Consider the 4 samples each having 2 features given below. Apply PCA to transform the data into k-dimensional space. Samples: $\begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix}$	K3	CO5	1.1.1 2.1.3 2.4.1 13.3.1
(Or)				
9.	Consider the given samples with two classes: W1 and W2. Apply LDA to transform the data into k-dimensional space. W1: $\begin{pmatrix} 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 7 \\ 5 \end{pmatrix}$ W2: $\begin{pmatrix} 9 \\ 4 \end{pmatrix}, \begin{pmatrix} 10 \\ 1 \end{pmatrix}, \begin{pmatrix} 12 \\ 3 \end{pmatrix}, \begin{pmatrix} 13 \\ 6 \end{pmatrix}$	K3	CO5	1.1.1 2.1.3 2.4.1 13.3.1
10.	Construct the K - Dimensional Tree and perform space splitting for the six 2-dimensional data points: (30,40), (5,25), (10,12), (70,70), (50,30), (35,45). Identify and specify the cutting dimension at each level of the k-d tree.	K3	CO3	1.1.1 2.1.3 2.4.1 13.3.1

	Search for the data point (50,30) and elucidate the process of searching the data point. [Marks: K -D Tree - 4, Space splitting - 4, Searching - 4]																																							
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11.	Consider the dataset given below and construct the decision tree model using information gain with entropy as an impurity metric. Identify the root node and expand one of the branches of the tree till the leaf node to identify the class label. Finally, identify and write the logical expression for the constructed sub tree from root node to the leaf. [Marks: Tree- 10, logical expression - 2] <table border="1"><thead><tr><th>Height</th><th>Hair</th><th>Eyes</th><th>Attractive?</th></tr></thead><tbody><tr><td>Small</td><td>Blonde</td><td>Brown</td><td>No</td></tr><tr><td>Tall</td><td>Dark</td><td>Brown</td><td>No</td></tr><tr><td>Tall</td><td>Blonde</td><td>Blue</td><td>Yes</td></tr><tr><td>Tall</td><td>Dark</td><td>Blue</td><td>No</td></tr><tr><td>Small</td><td>Dark</td><td>Blue</td><td>No</td></tr><tr><td>Tall</td><td>Red</td><td>Blue</td><td>Yes</td></tr><tr><td>Tall</td><td>Blonde</td><td>Brown</td><td>No</td></tr><tr><td>Small</td><td>Blonde</td><td>Blue</td><td>Yes</td></tr></tbody></table>	Height	Hair	Eyes	Attractive?	Small	Blonde	Brown	No	Tall	Dark	Brown	No	Tall	Blonde	Blue	Yes	Tall	Dark	Blue	No	Small	Dark	Blue	No	Tall	Red	Blue	Yes	Tall	Blonde	Brown	No	Small	Blonde	Blue	Yes	K3	CO3	1.1.1 2.1.3 2.4.1 13.3.1
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